

COMP 132 (01) ADVANCED PROGRAMMING

Spring 2025

1. Course Information

Instructor:	Öznur Özkasap, oozkasap@ku.edu.tr
KU Credits:	3.00
ECTS Credits:	6.00
Prerequisite(s):	Prerequisites: COMP 100 consent of the instructor
Class Location & Meeting Times:	SNA B152 - Monday, Wednesday 14:30-15:40
PS (Yes/No):	No
DS (Yes/No):	No
Lab (Yes/No):	Yes
Language of Instruction:	English
Office Hours:	Monday 11:30 - 12:30 (or by appointment)

2. Course Description

Object oriented programming using Java. Data types, expressions, control statements, strings, arrays. Classes, objects, methods, overloading, variable scope, memory. Recursion. Inheritance, polymorphism, abstract classes, interfaces, nested classes, anonymous classes. Exception handling. Strings and regular expressions. File I/O. Generic collections. Generic classes and methods. Lambdas and streams. Event-driven programming. Multithreading.

3. Course Overview**4. Course Learning Outcomes (CLOs):**

CLO #	Upon successful completion of this course, students will be able to...
1	ain extensive knowledge and understanding of advanced object-oriented programming concepts for software development
2	Apply object-oriented programming techniques such as inheritance, polymorphism, interfaces and generics
3	Design and develop software for real-life problems using advanced programming concepts

5. Assessment Methods

Method	Description	Weight %
Laboratory	Programming Labs	25.00
Midterm Exam	Midterm Exam and In-class Quizzes (Written)	20.00
Participation	Attendance and Interactive Questions	10.00
Project	Programming Project	15.00
Final Exam	Final Exam (Written and Programming)	30.00
Total:		100.00

6. Instructional Material and Learning Resources

- Java How to Program, Edition: 11
Author: Deitel
Publisher: Pearson
Material Type: Textbook
Material Status: Required
- sites.google.com/ku.edu.tr/comp132
Material Type: Website
Material Status: Required
- Active Use of Course Page on KUHUB Learn: <https://learn.hub.ku.edu.tr/>
- KOLT Tutoring: <http://kolt.ku.edu.tr/>

7. Course Schedule

Meeting Times	Subject
1	Please see the course site for details of the course schedule.

8. Student Code of Conduct and Academic Grievance Procedure

[Student Code of Conduct](#)

[Statement on Academic Honesty with Emphasis on Plagiarism](#)

[Academic Grievance Procedure](#)

9. Course Policies

Attendance Policy: Random attendance is taken in lectures. If your attendance percentage to lectures is below 65 %, you will fail this class (letter grade of F) regardless of your other grades. Grading policy: If your Lab average is below 40 (out of 100) you will fail this class (letter grade of F) regardless of your other grades.

10. Other

Grading policy:

- Programming Laboratory:
- The Laboratory is an important part of the course and it is mandatory. 100 minute lab sessions are continuous and is considered one lab. Your lab work will be graded during the lab. There will be around 8 graded lab sessions. Only the official health reports will be accepted as excuses, and only one lab can be excused. If you attend less than half of a lab or if you attend late, your lab will not be graded. There's no makeup for labs. . See the course web site for details.

If your Lab average is below 40 (out of 100) you will fail this class (letter grade of F) regardless of your final exam grade.

- Midterm Exam: Written Questions.
- In Class Quizzes:
These are short tests. They are part of the class participation. No makeup.
- Term Project (programming):
There will be one programming project in the second half of the course.
- Final Exam:
Written and Programming Questions.

Writing programs is the only way to learn how to program, therefore the lab/homework assignments and

programming part of the exams are very important learning tools and constitute a significant portion of your grade. The problems in lectures, labs, homeworks and exams would be similar in nature, thus lecture and lab attendance and the time you spend on homeworks and problems would have a large impact on your performance.

Attendance Policy: Random attendance is taken in lectures. If your attendance percentage to lectures is below 65 %, you will fail this class (letter grade of F) regardless of your other grades.

Any duplication in any programming assignments or exams will be regarded as cheating and will be prosecuted to the fullest extent allowed by university policy regarding academic dishonesty.

Text Book:

We will closely follow the required textbook.

Tools, infrastructure:

We use Eclipse Integrated Development Environment (IDE) for Java (8 or newer) in the lecture and labs, and github classroom. It is strongly recommended that you install eclipse/Java on your personal computer. IT will help in case you need help with the installation. See instructions in the course web page for installing these tools. Familiarize yourself with them by going through the tutorials installed with the tool as soon as possible.

Schedule and Subjects (During the course, slight changes are possible)

- Introduction and Java Overview: classes, objects, packages
- Control Statements, Expressions
- Methods - a deeper look: scope, overloading, call stack, parameter passing, recursion
- Arrays, Multidimensional Arrays, ArrayList
- Object Oriented Programming, Deeper look into Classes
- Inheritance
- Polymorphism: abstract classes, interfaces, type hierarchies
- Collections framework (Generic Collections)
- Exception handling
- Java GUI Programming - Event Driven Programming
- Strings, Regular Expressions
- Files, I/O Streams
- Generics: Generic methods, interfaces and classes
- Functional Programming with Java - Lambdas and Streams
- Multithreading, Concurrency